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Salary Prediction in the Tech Industry

STAT 155 Final
Presentation

Introduction

Research Question:

- ❖ Can we predict a professional's salary in the tech industry based on their job title, experience, location, and company attributes?

Motivation & Context:

- ❖ I'm a student entering the job market and curious about what drives salary differences.
- ❖ I'm interested in data analysis/data science and wanted to explore real-world salary patterns in tech.
- ❖ Salary transparency is a growing concern in today's job landscape.
- ❖ Understanding these factors can help job seekers make smarter, more informed career decision.

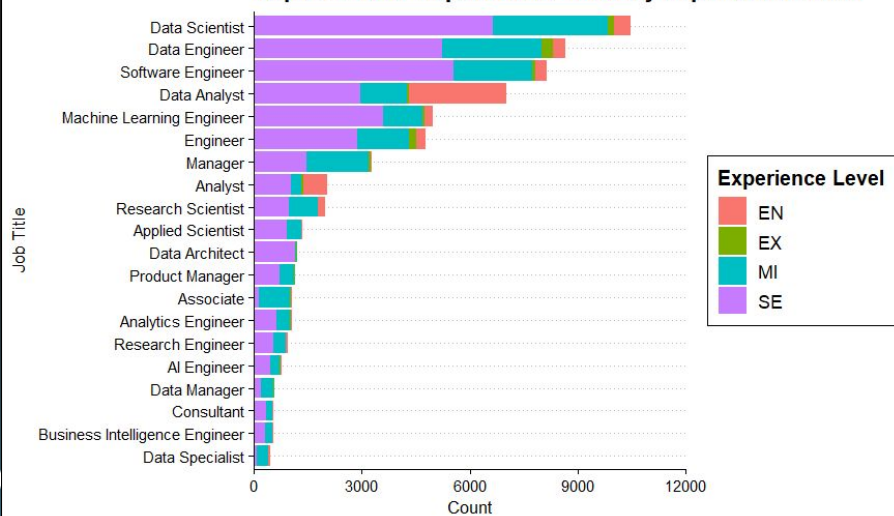
Data Wrangling

- ❖ Data Source: aijobs.net - Open Tech Salary Survey
- ❖ Format: Downloadable CSV; already organized and cleaned
- ❖ Over 100,000 observations and 11 features
 - Experience level, job title, employment type, company location, etc.
 - Excluded salary and currency because there is already a column for salary in USD.
- ❖ Data Split: 70% training, 15% validation, 15% testing
- ❖ Processing: Converted categorical columns using target encoding

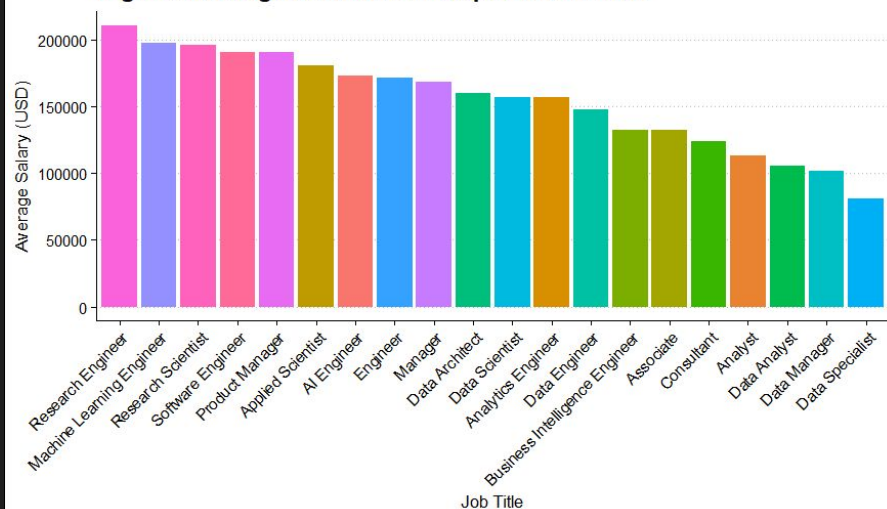
Exploratory Data Analysis

- ❖ The main predictors/covariates are:
 - Experience Level (4 levels)
 - Entry-level, Mid-level, Senior-level, Executive-level
 - Employment Type (4 levels)
 - Part time, Full time, Contract, Freelance
 - Job Title
 - Remote Ratio (3 levels)
 - 0 (less than 20%), 50, 100 (more than 80%)
 - Company's Location
 - Employee's Residence
 - Company Size (3 levels)
 - Small (<50), Medium (50-250), Large (>250)
- ❖ Target Variables: Salary (USD)

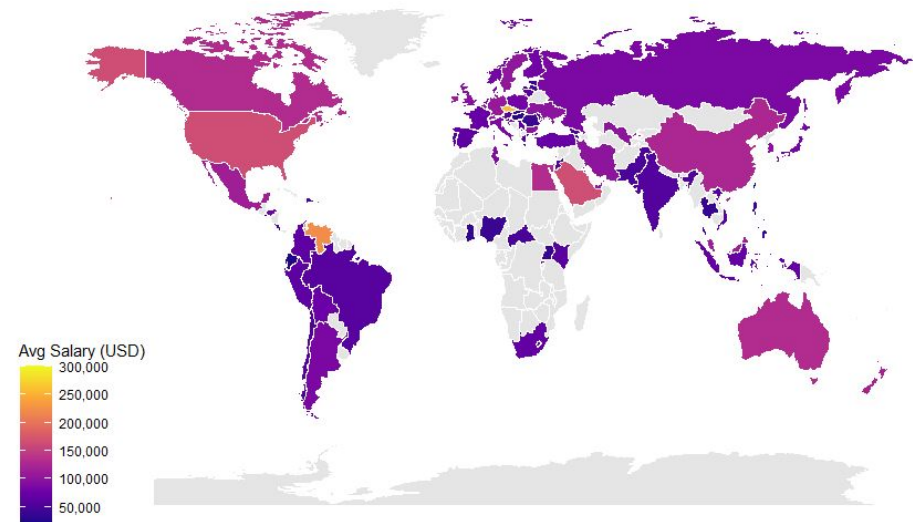
Top 20 Most Frequent Job Titles by Experience Level



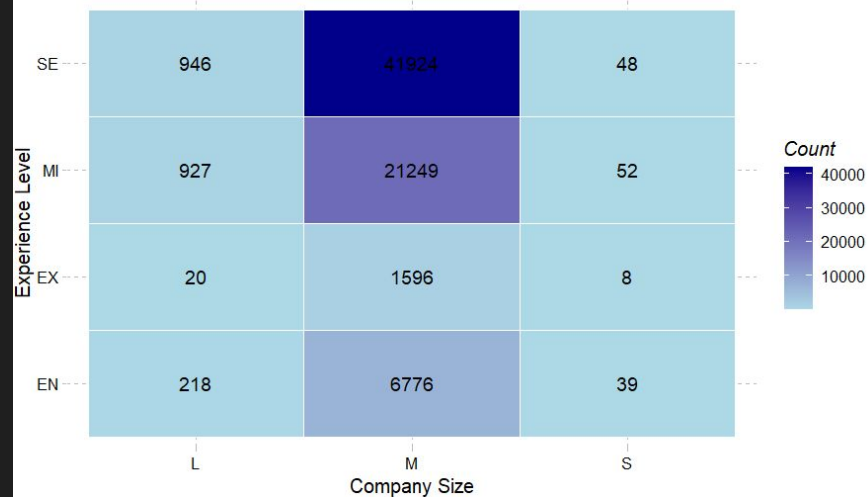
Highest Average Salaries for Frequent Job Titles



Global Distribution of Average Tech Salaries by Employee Residence



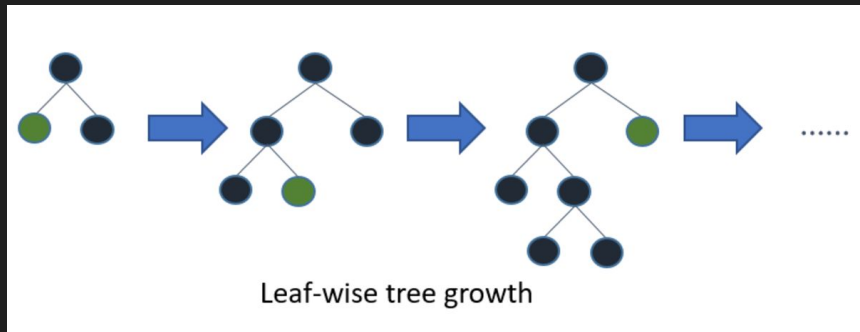
Experience Level by Company Size



Modeling – Light Gradient Boosting Machine

- ❖ A fast, high -performance machine learning framework based on decision tree algorithms, designed for:
 - Large datasets
 - High Dimensional data
 - Speed and Efficiency
- ❖ Builds an ensemble of decision trees, where each tree tries to correct the errors of the previous one
- ❖ Optimizes a loss function (like RMSE for regression) using gradient boosting
- ❖ Uses leaf-wise tree growth, which captures complex patterns better than traditional level-wise methods
- ❖ Model Parameters:
 - Number of Leaves: 31 - controls tree complexity
 - Learning Rate: 0.05 - how fast the model learns
 - Feature Fraction: 0.9 - randomly selects 90% of features to use during training to prevent overfitting
 - Bagging Fraction: 0.8 - uses 80% of data each round (random sampling)
 - Bagging Frequency: 5 - resample data every 5 rounds
 - Number of iterations: 1000 - number of boosting rounds

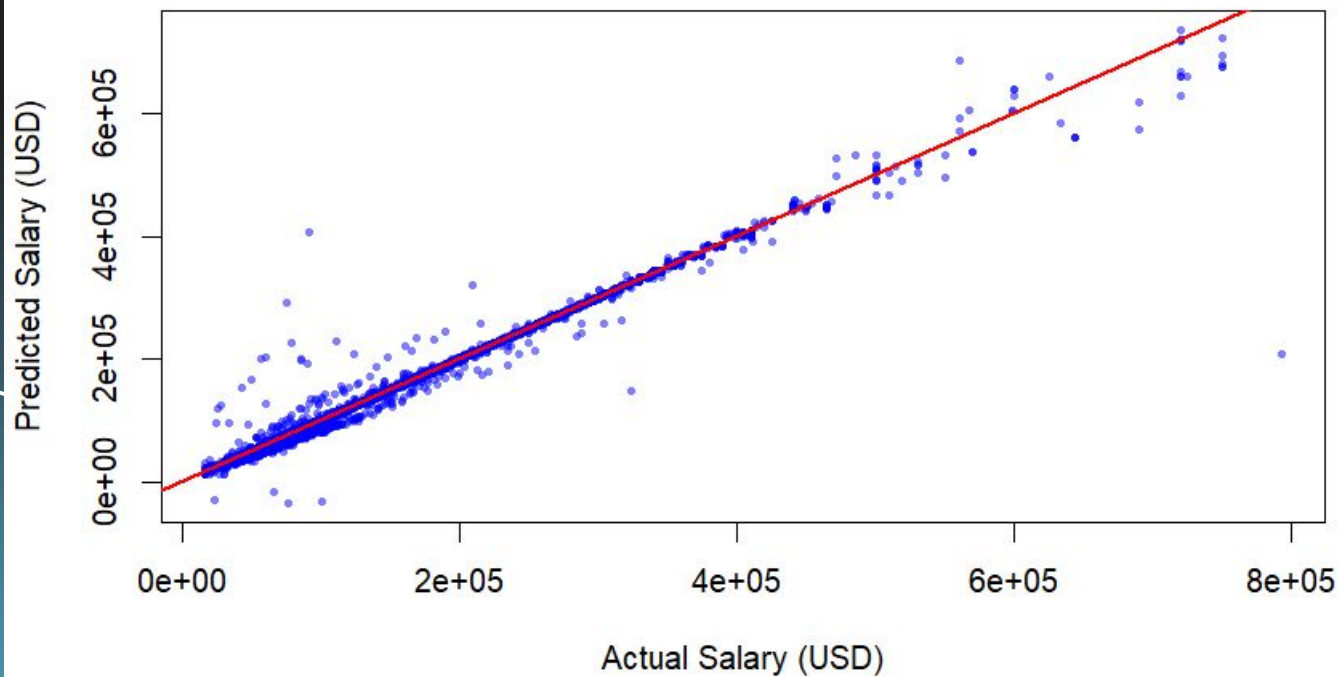
Modeling – Light Gradient Boosting Machine



- ❖ RMSE was used to evaluate the performance of the model
 - Measures how far predictions are from the actual values
- ❖ Training RMSE: 3,370
- ❖ Validation RMSE: 9,246
- ❖ Testing RMSE: 8,274

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$

Predicted vs Actual Salary



Monte Carlo Simulation

- ❖ Goal: Tune LLightGBM parameters effectively by simulating smaller, realistic subsets
- ❖ Simulation Design:
 - Ran 10 simulations, each with 500 observations
 - Randomly sampled:
 - 15 countries (company location, employee residence)
 - Top 20 most frequent job titles
 - Experience level, company size, employment type, remote ratio
 - Categorical variables converted to numeric rankings for modeling

Salary Formula

$$Y_i = \beta_0 + \beta_1 J_i + \beta_2 E_i + \beta_3 C_i + \beta_4 T_i + \beta_5 \left(\frac{R_i}{100} \right) + \varepsilon_i$$

Coefficients

- $\beta_0 = 0$: Intercept
- $\beta_1 = 0.7$: Job title salary effect
- $\beta_2 = 3000$: Experience level (ranked factor: EN = 1, MI = 2, SE = 3, EX = 4)
- $\beta_3 = 1500$: Company size (ranked: S = 1, M = 2, L = 3)
- $\beta_4 = 1000$: Employment type (ranked: FL, CT, PT, FT)
- $\beta_5 = 500$: Remote work percentage effect
- $\varepsilon_i \sim \mathcal{N}(0, 10,000^2)$: Normally distributed noise

Monte Carlo Simulation

- ❖ Parameter Grid
 - Number of Leaves: [15, 31, 63]
 - Learning Rate: [0.01, 0.05]
 - Feature Fraction: [0.7, 0.8, 0.9]
 - Bagging Fraction: [0.6, 0.8, 1.0]
 - Number of Iterations: [500, 1000]
 - Bagging Frequency: [1, 5]
- ❖ Best Model Parameters:
 - Number of Leaves: 15, 31
 - Learning Rate: 0.05
 - Feature Fraction: 0.8, 0..9
 - Bagging Fraction: 0.6
 - Number of Iterations: 100
 - Bagging Frequency: 5
 - Mean RMSE: 10,160.38

Final Answer to Research Question

- ❖ Research Question:
 - Can we predict a professional's salary in the tech industry based on their job title, experience, location, and company attributes?
- ❖ Short Answer:
 - Partially, my LightGBM model performed reasonably well on training data but had a testing RMSE of about \$8,274, meaning average predictions can still be off by a significant margin
- ❖ Key Predictive Features:
 - Job title, experience, level, company size, and employee residence had the most predictive power

Reflection on Process and Reproducibility

❖ What I learned:

- Data Cleaning and encoding decisions matter a lot (e.g. ranking factors like experience, target encoding)
- Model performance varies significantly depending on parameter tuning and the sample used
- Creating synthetic data let me explore model behavior under controlled setting

❖ Classmate Feedback:

- Seeing how others reproduced with my work reminded me of the importance of clear code and documentation

Ethical Implications

- ❖ Bias and Fairness:
 - If the model underpredicts salaries for certain groups (e.g. based on country, job title) it could reinforce pay disparities if using in hiring tools
- ❖ Transparency:
 - Encoding categorical variables as ranked numbers is a modeling choice many may not reflect reality
- ❖ Impact:
 - Salary predictions might be used in applications like job offer platforms or compensation benchmarks
 - Errors in model could lead to undervaluing someone's worth